

Solvency Field Theory V13.3

The First Locally Real Atom

Stability Band, Effective Proton Source, Directional Coulomb Well, Electron Support Balance, and Hydrogen Assembly

SFT V13.3 assembles the first locally real atom at leading nonrelativistic order.

Frozen Atom Stack

persistent builders : $e^-, u, d,$
effective proton : $uud \rightarrow p^+$ (interior guarded),
directional well : $V(r) = q_1 q_2 \frac{\alpha \hbar c}{r},$
support balance : $E(r) = \frac{\hbar^2}{2\mu r^2} - \frac{\alpha \hbar c}{r},$
hydrogen : $p^+ + e^- \rightarrow H^0.$

Frozen Numerical Anchors

Quantity	Value	Status
Reduced-mass Bohr radius	$a_\mu \approx 52.9465 \text{ pm}$	recovered with α input
Hydrogen binding	$E_{\text{bind}} \approx 13.5983 \text{ eV}$	recovered with α input
Leading hydrogen mass	$m_H c^2 \approx 938.78306665 \text{ MeV}$	leading assembly
Scale separation	$r_p/a_\mu \approx 1.586 \times 10^{-5}$	point-source pass

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1 Abstract

SFT V13.3 freezes the first-atom construction of Solvency Field Theory. The central result is that the R251–R256 chain assembles leading-order hydrogen from locally real settlement ingredients: the persistent charged builder band supplies the electron and quark seeds; R252–R253 justify a guarded effective proton source; R254A produces a Coulomb-shape exterior from conserved directional settlement flux; R255A balances the electron's $-1/r$ directional attraction against a $+1/r^2$ closure-support cost; and R256A combines the pieces into a neutral bound atom. With α used as an input normalization, the construction recovers the reduced-mass Bohr radius $a_\mu \approx 52.9465$ pm and binding $E_{\text{bind}} \approx 13.5983$ eV. This is not a derivation of α , not a full proton derivation, not a full hydrogen spectrum, and not chemistry. It is the leading nonrelativistic hydrogen assembly gate.

2 Executive Claim and Boundary

V13.3 First-Atom Claim

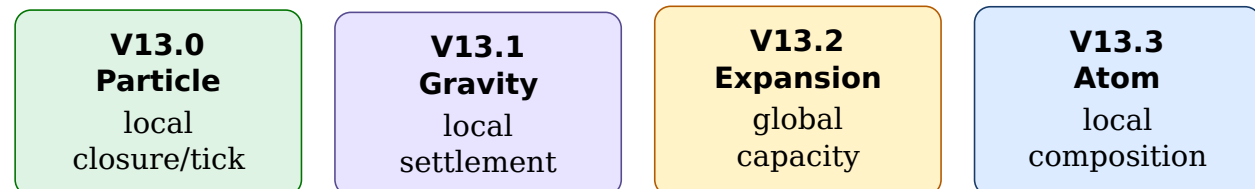
A guarded effective proton provides a $+1$ directional settlement source; an electron supplies a -1 persistent closure; directional attraction gives a $1/r$ well; electron support maintenance gives a $1/r^2$ collapse barrier; the balance gives the Bohr radius and 13.6 eV binding, forming neutral hydrogen.

Immediate Claim Boundary

V13.3 freezes leading-order hydrogen assembly. It does not derive α , does not derive the proton baryon junction or confinement kernel, does not solve the full hadron spectrum, does not derive fine structure, hyperfine splitting, Lamb shift, relativistic corrections, or chemistry.

3 Continuity Map: From Particle to Atom

V13.3 is the particle-to-atom integration branch. It continues the V13 sequence:



V13.3 Freeze Sentence

SFT V13.3 assembles the first locally real atom at leading nonrelativistic order, using α as an input and keeping the proton interior guarded.

4 Native Ontology

V13.3 does not treat every Standard Model label as equally fundamental. It follows the particle reality sorting used in the V13 program: persistent builder modes supply the stable

structures of ordinary matter, transient modes are visitors or resonances, and virtual/hypothetical sectors are not treated as persistent physical builders in native SFT.

The native atom-building objects are:

Object	Native role
Persistent builder	A locally real stable/effectively stable support mode.
Effective proton	A guarded confined settlement structure with exterior charge +1.
Directional settlement	Charge-address compatibility whose conserved exterior flux gives Coulomb shape.
Electron support domain	The electron's closure-maintenance region; localization costs scale as $1/r^2$.
Hydrogen atom	Neutral bound composite settlement structure, $p^+ + e^-$.

5 R251-R257 Gate Rollup

Gate	Name	Verdict	Core result
R251A	Stability Band / Minimum Viable Compactness	PASS	Stable charged builder band identified: electron/up/down, 0.511–4.7 MeV, factor ~ 9.2 ; minimum viable compactness framed as species-map target.
R252A	Quark Confinement Toy Gate	PARTIAL	Raw G_A compactness at 1 fm fails; nuclear settlement support/tension target ~ 0.929 GeV/fm identified.
R252B	L_{eff} Settlement-Overlap Toy	PARTIAL	L_{eff} -compounded overlap can bridge to proton field-energy target with modest u ; three-link $\hbar c/\text{fm}$ base needs $u \sim 0.104$.
R253A	Proton Settlement-Energy Requirement	PARTIAL	uud seed ~ 9.1 MeV; proton ~ 938.27 MeV; field target ~ 929.17 MeV = $4.71 \hbar c/\text{fm}$ total.
R253B	Three-Body Overlap / Pion Ratio Toy	PARTIAL	Proton/pion ratio requires $\eta_3 \sim 2.2$ – 2.4 beyond three independent links; extra L_{eff} completion $\Delta u \sim 0.045$ – 0.049 .
R254A	Directional Settlement / Coulomb Well	PASS	Effective +1 proton directional flux in 3D gives field $\sim 1/r^2$ and potential $\sim 1/r$; electron-proton sign attractive.
R255A	Electron Orbital Equilibrium	PASS	$-1/r$ attraction plus $+1/r^2$ support cost gives finite minimum; $a_\mu \sim 52.9465$ pm and binding ~ 13.5983 eV with α input.

Gate	Name	Verdict	Core result
R256A	Hydrogen Atom Assembly	PASS	Effective proton +1 and electron −1 form neutral bound settlement structure; leading hydrogen assembled.
R257A	First Atom Freeze	PASS	R251-R256 chain frozen as SFT V13.3 first locally real atom construction.

6 Frozen Equation Stack

Native Atom Equation Spine

$$r_{\text{spin}} = \frac{\hbar}{2mc}, \quad (1)$$

$$E_{\text{field},p} = m_p c^2 - (2m_u + m_d)c^2, \quad (2)$$

$$V(r) = q_1 q_2 \frac{\alpha \hbar c}{r}, \quad (3)$$

$$K(r) = \frac{\hbar^2}{2\mu r^2}, \quad (4)$$

$$E(r) = \frac{\hbar^2}{2\mu r^2} - \frac{\alpha \hbar c}{r}, \quad (5)$$

$$a_\mu = \frac{\hbar}{\mu c \alpha}, \quad (6)$$

$$E_{\text{bind}} = \frac{\mu c^2 \alpha^2}{2}, \quad (7)$$

$$q_p + q_e = +1 - 1 = 0. \quad (8)$$

7 R251A: Stable Charged Builder Band

The stable charged matter universe is built from a narrow ground-state band:

Builder	Charge	Mass scale	$r_{\text{spin}} = \hbar/(2mc)$
Electron	−1	0.51099895 MeV	193.08 fm
Up quark seed	+2/3	~ 2.2 MeV	44.85 fm
Down quark seed	−1/3	~ 4.7 MeV	20.99 fm

The band spans only a factor

$$\frac{m_d}{m_e} \approx 9.20.$$

The sharpened V13.3 target is not to derive a 61-particle zoo at once. It is:

Derive the minimum viable compactness for each persistent charge/spin/confinement family.

8 R252-R253: Guarded Effective Proton

The proton interior is not fully derived in V13.3. It is treated as a guarded effective source for atom assembly.

The seed accounting is:

$$\begin{aligned} 2m_u + m_d &\approx 9.1 \text{ MeV}, \\ m_p &\approx 938.2720813 \text{ MeV}, \\ E_{\text{field},p} &\approx 929.1720813 \text{ MeV}. \end{aligned}$$

Thus roughly 99% of the proton mass-energy is not in the simple current-quark seed accounting. It is assigned to guarded confinement/field/settlement energy.

Nuclear Support Scale

R252-R253 found that the proton field-energy target is nuclear-natural, not absurd, once measured in $\hbar c/\text{fm}$ units:

$$\frac{E_{\text{field},p}}{\hbar c/\text{fm}} \approx 4.71, \quad \frac{E_{\text{field},p}/3}{\hbar c/\text{fm}} \approx 1.57.$$

A single 1 fm support gives a toy target $\sigma \sim 0.929 \text{ GeV}/\text{fm}$. A Y or Δ support changes the required tension, so topology remains open.

Proton Guardrail

The proton is used as an effective +1 source for atomic assembly. The baryon junction, color/confinement topology, and full proton mass derivation remain open.

9 R254A: Directional Settlement Well

Given an effective proton charge source, R254A establishes the exterior shape.

Assume a conserved directional settlement flux Φ_Q through enclosing spheres. In three spatial dimensions:

$$A(r) = 4\pi r^2.$$

Constant flux implies field density:

$$D(r) \propto \frac{1}{r^2}.$$

The potential is the radial integral of the field:

$$V(r) \propto \frac{1}{r}.$$

With the standard-readable electromagnetic normalization:

$$V(r) = q_1 q_2 \frac{\alpha \hbar c}{r}.$$

For proton-electron:

$$q_p q_e = (+1)(-1) = -1,$$

so the well is attractive.

R254A Preservation

Conserved directional settlement flux in 3D gives field $\sim 1/r^2$ and potential $\sim 1/r$. Alpha sets the normalization and is not derived in R254A.

10 R255A: Electron Support Balance

R255A adds the electron closure/support cost to the directional well:

$$E(r) = \frac{\hbar^2}{2\mu r^2} - \frac{\alpha \hbar c}{r}.$$

The $-1/r$ term pulls inward. The $+1/r^2$ term prevents collapse because localizing the electron support domain too tightly raises closure-maintenance cost.

Minimizing:

$$\frac{dE}{dr} = -\frac{\hbar^2}{\mu r^3} + \frac{\alpha \hbar c}{r^2} = 0$$

gives:

$$a_\mu = \frac{\hbar}{\mu c \alpha}.$$

At the minimum:

$$\begin{aligned} K &= +\frac{\mu c^2 \alpha^2}{2}, \\ V &= -\mu c^2 \alpha^2, \\ E &= -\frac{\mu c^2 \alpha^2}{2}. \end{aligned}$$

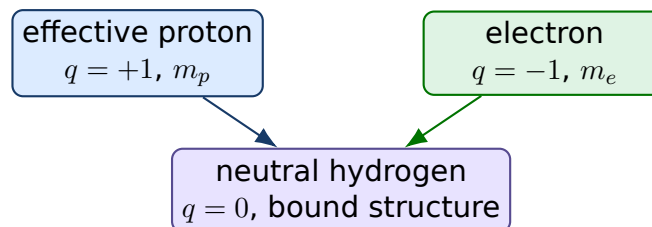
The virial relation is:

$$2K + V = 0.$$

Quantity	Infinite-proton approximation	Reduced-mass hydrogen
Radius	$a_0 \approx 52.9177 \text{ pm}$	$a_\mu \approx 52.9465 \text{ pm}$
Binding	13.6057 eV	13.5983 eV

11 R256A: Hydrogen Assembly

The atom is assembled as:



directional attraction + support balance

The leading assembly gives:

$$\begin{aligned} q_H &= +1 - 1 = 0, \\ a_\mu &\approx 52.9465 \text{ pm}, \\ E_{\text{bind}} &\approx 13.5983 \text{ eV}, \\ m_H c^2 &\approx 938.78306665 \text{ MeV}. \end{aligned}$$

12 Scale Hierarchy

The scale hierarchy passes the point-source and support-domain checks:

Scale	Value	Relative to a_μ	Meaning
Nuclear support	$\sim 1 \text{ fm}$	1.89×10^{-5}	proton interior scale
Proton radius ref.	0.84 fm	1.59×10^{-5}	finite-size check
Electron r_{spin}	193.08 fm	3.65×10^{-3}	internal electron closure
Hydrogen a_μ	52946.5 fm	1	orbital support radius

Thus the proton interior is negligible for the leading electronic shell, and the electron internal closure scale is much smaller than the atomic support domain:

$$\frac{a_\mu}{r_{\text{spin},e}} \approx 274.2.$$

13 Claim Boundary

Status	Claim
Promoted	Leading-order hydrogen assembly from settlement mechanics, with α input and proton interior guarded.
Promoted	Coulomb-shape directional exterior: field $\sim 1/r^2$, potential $\sim 1/r$.
Promoted	Electron support balance: $-1/r$ attraction plus $+1/r^2$ support cost gives Bohr radius and binding.
Open	Derivation of α as directional-to-scalar settlement ratio.
Open	Proton baryon junction, color/confinement topology, and full proton mass.
Open	Full hydrogen spectrum, fine structure, hyperfine splitting, Lamb shift, and relativistic corrections.
Deferred	Molecules, chemical bonds, and chemistry.
Blocked wording	Do not say “the strong force is just gravity.” Use: strong/nuclear binding is a different compounded settlement-support regime, not raw long-range gravity compactness.

14 Next Tracks

1. **R258A - V13.3 First-Atom Paper.** Write the polished manuscript from this freeze.

2. **R257B - Alpha Derivation Target Gate.** Define the directional-to-scalar settlement ratio target for α .
3. **R257C - Hydrogen Spectrum Extension.** Extend from ground-state Bohr/Rydberg to n -level spectrum, then guard fine structure.
4. **R253C - Baryon Junction / Triple-Overlap Kernel.** Return to the proton interior and nuclear topology.
5. **R259A - H₂ / Chemistry Toy Gate.** Begin multi-atom settlement bonding only after hydrogen freeze is stable.

15 Final Preservation Statement

R257A Preservation

R257A freezes SFT V13.3 as the first locally real atom construction. The R251–R256 chain identifies the persistent charged builders, treats the proton as a guarded effective $+1$ confined settlement source, derives the Coulomb-shape directional exterior, balances the electron's $-1/r$ attraction against $+1/r^2$ closure support cost, and assembles neutral hydrogen with $a_\mu \approx 52.9465$ pm and binding ≈ 13.5983 eV using α as input. The result is leading nonrelativistic hydrogen, not α , not the proton interior, not full spectroscopy, and not chemistry.

A Appendix: Frozen Numeric Anchors

Quantity	Value	Unit	Role
Electron mass	0.51099895	MeV	persistent charged builder
Up quark seed scale	2.2	MeV	$+2/3$ confined seed
Down quark seed scale	4.7	MeV	$-1/3$ confined seed
uud seed	9.1	MeV	proton seed accounting
Proton mass	938.2720813	MeV	effective proton source
Proton field target	929.1720813	MeV	guarded confinement energy target
σ single 1 fm target	0.929	GeV/fm	toy nuclear support target
L_{eff}	9.097814727	dimensionless	compounding parameter
α	0.00729735256	dimensionless	input normalization
μ/m_e	0.9994556794	dimensionless	reduced-mass correction
a_μ	52.9465	pm	assembled radius

Quantity	Value	Unit	Role
E_{bind}	13.5983	eV	assembled binding
$m_H c^2$	938.78306665	MeV	leading atom mass
r_p/a_μ	1.586×10^{-5}	dimensionless	scale separation
$a_\mu/r_{\text{spin},e}$	274.22	dimensionless	orbital/internal separation

B Appendix: Equation Index

Label	Equation	Meaning
Builder radius	$r_{\text{spin}} = \hbar/(2mc)$	persistent internal closure scale
Proton field target	$E_{\text{field},p} = m_p c^2 - (2m_u + m_d)c^2$	guarded proton interior target
Directional well	$V(r) = q_1 q_2 \alpha \hbar c / r$	Coulomb translation of directional settlement
Support cost	$K(r) = \hbar^2/(2\mu r^2)$	electron localization/support-domain cost
Balance functional	$E(r) = \hbar^2/(2\mu r^2) - \alpha \hbar c / r$	leading hydrogen energy functional
Bohr radius	$a_\mu = \hbar/(\mu c \alpha)$	reduced-mass orbital radius
Binding	$E_{\text{bind}} = \mu c^2 \alpha^2 / 2$	leading hydrogen binding
Neutrality	$q_p + q_e = 0$	neutral atom closure